

Embedded systems can be applied to a broad range of electronic medical equipment such as X-ray machines, ultrasonic diagnosis apparatus, computer tomographs, heart pacemakers and patient monitoring systems.

As medical designs continue to shrink in size, the design challenges and limitations within medical electronics become more pronounced. Miniaturisation continues to have a huge impact on medical electronics because it drives portability and accessibility. Areas like telemedicine and body area networks are completely dependent on the miniaturisation of medical devices. It gives the ability to save and improve lives worldwide because the right equipment can be easily transported whenever and wherever it is needed.

While up-integration can increase the functionality achieved with a single integrated circuit (IC), redundant support for critical functions becomes less feasible should that IC fail. Similarly, the number of PC board layers needed to route the traces for the device can increase with smaller bond pitches and tiny wafer chip-scale packages. This can require hidden vias on the board, which create through-hole connections that cannot be visually inspected. The

good news is that these tradeoffs can be compensated for, if understood and defined early enough in the development process.

Looking into the future

The worldwide medical electronics market grew an estimated twelve per cent in total sales from \$139 billion in 2010 to \$156 billion in 2011. In the next five years, the market is expected to grow nine per cent a year, reaching \$243.2 billion by 2016. Over the last decade, interesting trends have emerged. Rapidly ageing population in developed countries and the need for basic healthcare in developing countries is quickly becoming a challenge for the healthcare industry. This gives enormous opportunities in medical electronics.

Along with the increasing application of network technologies and mobile terminals in the medical sector, the future of medical electronics will be paved with technologies that allow portability, connectivity and data security. Leveraging these technologies, systems will move quickly from the hospital environment to the home, enabling caregivers from doctors to family



Onyx-II (model 9560) fingertip pulse oximeter

members to monitor patients' biological trends and events. For this to happen, secure infrastructure and feature sets in the monitoring systems are the prerequisites.

Knowledge of analogue and processing solutions, dedication to reliability and continued investments in the market will put companies in a leadership position to help manufacturers of medical devices optimise their designs now and in the future. In order to meet the miniaturisation, high integration as well as low-power consumption requirements of portable medical devices, semiconductor vendors are increasing their investment focus on the research and development of electronic components for portable medical devices.

The adoption of electronics technology, however, cannot be achieved in isolation from other technologies like nano-technology, materials, power sources, sensors and micro-fluidics. ●

A new breed of engineers required

The research and development work of medical electronics engineers leads to the manufacturing of sophisticated diagnostic medical equipment needed to ensure good healthcare. Biomedical engineering combines the design and problem-solving skills of engineering with medical and biological sciences to improve healthcare diagnosis and treatment. Much of the work in biomedical engineering consists of research and development spanning a broad array of sub-fields.

The core healthcare science and research in medical sciences will have ever-increasing interface with technology areas. To meet these challenges, a new breed of medical professionals is required which is conversant with medical as well as engineering aspects. They will be able to fuse together the medical sciences with high-end technologies.

Medical electronics engineers carry out research along with life scientists, chemists and medical scientists to develop and evaluate systems and products such as bio-compatible prostheses (artificial devices that replace missing body parts), various diagnostic and therapeutic medical devices ranging from clinical equipment to micro-implants, common imaging equipment such as magnetic resonance imaging (MRI) and electroencephalogram (EEG), biotechnologies such as regenerative tissue growth, pharmaceutical drugs and biopharmaceuticals, medical information systems, and health management and care delivery systems.

Most engineers in this speciality need a sound background in another engineering speciality, such as mechanical or electronics engineering, in addition to specialised biomedical training. Some specialities within medical electronics engineering include biomaterials, biomechanics, medical imaging, rehabilitation engineering and orthopaedic engineering.

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